

■ B.TECH 1ST YEAR · 1ST SEMESTER · 2024–25

BEEE Formula Sheet

Basic Electrical & Electronics Engineering

⚡ UNIT I — DC & AC CIRCUITS

"Your last-day revision weapon — compact, complete, exam-ready"



DC Circuits

Ohm's Law · KCL · KVL
Series · Parallel · Superposition



AC Fundamentals

Waveform · RMS · Avg
Form Factor · Phasors



Power

P · Q · S · PF
Power Triangle



R-L-C Circuits

Reactance · Impedance
Phase Angles



30 Flashcards

Q&A format
Exam-focused



Exam Tips

Common mistakes
Quick revision page

⚡ DC CIRCUITS — FUNDAMENTALS

BASIC CIRCUIT ELEMENTS

● Resistance (R)

$$R = \rho L / A$$

- **Unit:** Ohm (Ω)
- ρ = resistivity ($\Omega \cdot m$)
- L = length (m), A = area (m^2)
- Opposes current flow
- Dissipates energy as heat

● Inductance (L)

$$V = L \cdot dI/dt$$

- **Unit:** Henry (H)
- Opposes change in current
- Stores energy in magnetic field
- $E = \frac{1}{2}LI^2$
- Acts as short at DC (steady state)

● Capacitance (C)

$$I = C \cdot dV/dt$$

- **Unit:** Farad (F)
- Opposes change in voltage
- Stores energy in electric field
- $E = \frac{1}{2}CV^2$
- Acts as open at DC (steady state)

OHM'S LAW

$$V = I \times R \quad | \quad I = V/R \quad | \quad R = V/I$$

$$P = VI = I^2R = V^2/R \quad [\text{Watts}]$$

$$G = 1/R \quad [\text{Siemens} - \text{Conductance}]$$

⚠ Limitations of Ohm's Law

- NOT valid for non-linear devices (diodes, transistors)
- NOT valid at very high temperatures
- NOT valid for unilateral elements
- Assumes constant temperature
- Valid only for metallic conductors

KIRCHHOFF'S LAWS

🔌 KCL — Kirchhoff's Current Law

$$\Sigma I_{in} = \Sigma I_{out} \quad | \quad \Sigma I = 0 \quad (\text{at node})$$

Statement: The algebraic sum of all currents meeting at a node is zero.

Based on: **Conservation of Charge**

Sign Convention: Current entering = +ve, leaving = -ve (or vice-versa, but stay consistent)

🔌 KVL — Kirchhoff's Voltage Law

$$\Sigma V = 0 \quad (\text{around any closed loop})$$

Statement: The algebraic sum of all voltages in a closed loop is zero.

Based on: **Conservation of Energy**

Sign Convention: Rise = +ve, Drop = -ve (follow loop direction)

SERIES & PARALLEL CIRCUITS

Parameter	Series Circuit	Parallel Circuit
Current (I)	$I = I_1 = I_2 = I_3 \quad (\text{same})$	$I = I_1 + I_2 + I_3$
Voltage (V)	$V = V_1 + V_2 + V_3$	$V = V_1 = V_2 = V_3 \quad (\text{same})$
Equivalent R	$R_{eq} = R_1 + R_2 + R_3$	$1/R_{eq} = 1/R_1 + 1/R_2 + 1/R_3$
For 2 resistors	$R_{eq} = R_1 + R_2$	$R_{eq} = R_1 R_2 / (R_1 + R_2)$
Power	$P = P_1 + P_2 + P_3$	$P = P_1 + P_2 + P_3$
Req vs R	$R_{eq} > \text{any individual R}$	$R_{eq} < \text{smallest R}$

VOLTAGE & CURRENT DIVISION RULES

⚡ Voltage Divider Rule (Series)

$$V_1 = V \times R_1 / (R_1 + R_2)$$

$$V_2 = V \times R_2 / (R_1 + R_2)$$

🔑 Voltage divides in proportion to resistance

🔄 Current Divider Rule (Parallel)

$$I_1 = I \times R_2 / (R_1 + R_2)$$

$$I_2 = I \times R_1 / (R_1 + R_2)$$

🔑 Current divides in inverse proportion to resistance

SUPERPOSITION THEOREM

 **Statement**

In a linear bilateral circuit with **multiple independent sources**, the response (voltage or current) in any element is the **algebraic sum of responses** caused by each independent source acting alone, with all other sources replaced by their internal resistances.

$$I_{total} = I_1 + I_2 + I_3 + \dots$$

Valid for: Linear, bilateral circuits only
NOT valid for: Power ($P = I^2R$ — non-linear)

 **Steps to Apply Superposition**

- Select one independent source, deactivate all others:
 - Voltage source → **Short Circuit** ($V=0$)
 - Current source → **Open Circuit** ($I=0$)
- Find response (I or V) due to the active source alone
- Repeat for each source
- Algebraically add all individual responses
- Note direction/polarity before adding

DC FORMULAS QUICK SUMMARY

Quantity	Formula	Unit	Notes
Voltage	$V = IR$	Volt (V)	Ohm's Law
Current	$I = V/R$	Ampere (A)	Ohm's Law
Resistance	$R = V/I = \rho L/A$	Ohm (Ω)	Material property
Power	$P = VI = I^2R = V^2/R$	Watt (W)	Always positive for R
Energy	$W = P \times t = VIt$	Joule (J) / kWh	1 kWh = 3.6 MJ
Conductance	$G = 1/R$	Siemens (S)	Reciprocal of R
Series Req	$R_1 + R_2 + \dots + R_n$	Ω	Larger than any R
Parallel Req	$1 / (1/R_1 + 1/R_2 + \dots + 1/R_n)$	Ω	Smaller than any R
Two Parallel R	$R_1 R_2 / (R_1 + R_2)$	Ω	Product over sum
Voltage Divider	$V_x = V \times R_x / Req$	V	Series only
Current Divider	$I_x = I \times R_{other} / Req$	A	Parallel only (2R)

DC VS AC — COMPARISON

Parameter	DC (Direct Current)	AC (Alternating Current)
Direction	Constant direction	Reverses periodically
Waveform	Straight line (constant)	Sinusoidal / varying
Frequency	0 Hz ($f = 0$)	50 Hz (India) / 60 Hz (USA)
Magnitude	Constant	Varies (peak, RMS, avg)
Generation	Batteries, rectifiers	Alternators, generators
Transmission	Not efficient long distance	Efficient (step up/down)
Passive elements	R only (L=short, C=open)	R, L, C all active
Power	$P = VI$	$P = VI \cdot \cos\phi$ (real power)
Safety	Safer at low voltage	More dangerous

SINUSOIDAL AC EQUATIONS

$$v(t) = V_m \sin(\omega t \pm \phi)$$

$$i(t) = I_m \sin(\omega t \pm \phi)$$

$$\omega = 2\pi f = 2\pi/T \quad [\text{rad/s}]$$

$$f = 1/T \quad T = 1/f \quad [\text{Hz, seconds}]$$

Waveform Parameters

V_m (or I_m)	Peak / Amplitude
T	Time Period (seconds)
f	Frequency (Hz)
ω	Angular frequency (rad/s)
ϕ	Phase angle (degrees/rad)
$+\phi$	Leading (ahead)
$-\phi$	Lagging (behind)

KEY WAVEFORM VALUES (SINUSOIDAL)

Parameter	Formula	Value (for sine)	Description
Peak Value (V_m)	V_m	V_m	Maximum amplitude
Peak-to-Peak	$V_{pp} = 2V_m$	$2V_m$	Total swing
Average Value	$V_{avg} = 2V_m/\pi$	$0.637 V_m$	Over half cycle ($ \sin $)
RMS Value	$V_{rms} = V_m/\sqrt{2}$	$0.707 V_m$	Effective / heating value
Form Factor (K_f)	$K_f = V_{rms}/V_{avg}$	1.11	$\pi/(2\sqrt{2}) \approx 1.11$
Peak/Crest Factor (K_p)	$K_p = V_m/V_{rms}$	$\sqrt{2} \approx 1.414$	Ratio of peak to RMS

Remember: Average value over a full cycle of a pure sine wave = 0. The $0.637V_m$ is for a **half-wave rectified** or $|\sin|$ waveform.

PHASE & PHASE DIFFERENCE

Phase Relationship

If $v = V_m \sin(\omega t + \phi_1)$ and $i = I_m \sin(\omega t + \phi_2)$

Phase difference = $\phi_1 - \phi_2$

- If $\phi_1 > \phi_2 \rightarrow v$ leads i
- If $\phi_1 < \phi_2 \rightarrow v$ lags i
- If $\phi_1 = \phi_2 \rightarrow$ in phase
- If $|\phi_1 - \phi_2| = 90^\circ \rightarrow$ in quadrature
- If $|\phi_1 - \phi_2| = 180^\circ \rightarrow$ in antiphase

Important Conversions

$$\omega = 2\pi f \quad [\text{rad/s}]$$

$$f = \omega/2\pi \quad [\text{Hz}]$$

$$T = 1/f = 2\pi/\omega \quad [\text{s}]$$

$$V_{rms} = V_m/\sqrt{2} \approx 0.707 V_m$$

$$I_{rms} = I_m/\sqrt{2} \approx 0.707 I_m$$

Indian supply: 230V RMS, 50 Hz $\rightarrow V_m = 230\sqrt{2} \approx 325$ V

PURE RESISTIVE, INDUCTIVE & CAPACITIVE CIRCUITS

Parameter	Pure Resistor (R)	Pure Inductor (L)	Pure Capacitor (C)
Voltage eq.	$v = V_m \sin(\omega t)$	$v = V_m \sin(\omega t)$	$v = V_m \sin(\omega t)$
Current eq.	$i = I_m \sin(\omega t)$	$i = I_m \sin(\omega t - 90^\circ)$	$i = I_m \sin(\omega t + 90^\circ)$
Phase angle (ϕ)	0° (in phase)	$+90^\circ$ (V leads I)	-90° (I leads V)
Impedance	$Z = R$	$Z = jX_L = j\omega L$	$Z = -jX_C = 1/j\omega C$
Reactance	—	$X_L = \omega L = 2\pi fL \text{ } [\Omega]$	$X_C = 1/\omega C = 1/2\pi fC \text{ } [\Omega]$
Current (I_m)	$I_m = V_m/R$	$I_m = V_m/X_L$	$I_m = V_m/X_C$
Power factor	1 (unity)	0 (lagging)	0 (leading)
Active Power	$P = V^2/R = I^2R$	$P = 0$ (no real power)	$P = 0$ (no real power)
Energy stored	Dissipated as heat	$E = \frac{1}{2}LI^2 \text{ (magnetic)}$	$E = \frac{1}{2}CV^2 \text{ (electric)}$
At $f \rightarrow 0$ (DC)	R (normal)	Short circuit ($X_L \rightarrow 0$)	Open circuit ($X_C \rightarrow \infty$)
At $f \rightarrow \infty$	R (normal)	Open circuit ($X_L \rightarrow \infty$)	Short circuit ($X_C \rightarrow 0$)

IMPEDANCE (Z) — GENERAL FORMULAS

$Z = R + jX$ [Complex impedance]

$|Z| = \sqrt{R^2 + X^2}$ [Magnitude — Ω]

$\phi = \tan^{-1}(X/R)$ [Phase angle]

R-L Series	$Z = R + jX_L = \sqrt{R^2 + X_L^2} \angle \tan^{-1}(X_L/R)$
R-C Series	$Z = R - jX_C = \sqrt{R^2 + X_C^2} \angle -\tan^{-1}(X_C/R)$
R-L-C Series	$Z = R + j(X_L - X_C) = \sqrt{R^2 + (X_L - X_C)^2}$

 **Phasor Diagram Rules**

- **Reference:** Current (I) is reference for series circuits
- **V_R:** In phase with I (horizontal)
- **V_L:** 90° ahead of I (vertical up)
- **V_C:** 90° behind I (vertical down)
- **V = $\sqrt{V_R^2 + (V_L - V_C)^2}$**
- If $X_L > X_C \rightarrow$ Inductive (lagging)
- If $X_C > X_L \rightarrow$ Capacitive (leading)
- If $X_L = X_C \rightarrow$ Resonance (unity PF)

SERIES RESONANCE

At resonance: $X_L = X_C$ | $f_r = 1/(2\pi\sqrt{LC})$ | $Z = R$ (minimum) | $I = \text{maximum}$

⚡ POWER IN AC CIRCUITS

TYPES OF POWER

● Active / Real Power (P)

$$P = VI \cos\phi$$

$$P = I^2 R = V^2 / R$$

- Unit: **Watt (W)**
- Actually consumed power
- Dissipated in resistance only
- Does useful work

● Reactive Power (Q)

$$Q = VI \sin\phi$$

$$Q = I^2 X = V^2 / X$$

- Unit: **VAR**
- Not consumed — exchanged
- Stored/released by L and C
- $Q_L = +ve$, $Q_C = -ve$

● Apparent Power (S)

$$S = VI = \sqrt{P^2 + Q^2}$$

$$S = I^2 Z = V^2 / Z$$

- Unit: **VA**
- Total power supplied
- Rating of equipment
- $S = \sqrt{P^2 + Q^2}$

POWER TRIANGLE & POWER FACTOR

📐 Power Triangle Relationships

$$S^2 = P^2 + Q^2$$

$$P = S \cos\phi \quad [\text{Active}]$$

$$Q = S \sin\phi \quad [\text{Reactive}]$$

$$\cos\phi = P/S = R/Z$$

$$\tan\phi = Q/P = X/R$$

🔑 S is hypotenuse, P is base, Q is perpendicular

⚡ Power Factor ($\cos\phi$)

$$PF = \cos\phi = P/S = R/Z$$

PF value	Circuit type	Nature
PF = 1	Pure R	Unity
PF = 0	Pure L or C	Zero
$0 < PF < 1$	R-L or R-C	0 to 1
PF lagging	Inductive load	I lags V
PF leading	Capacitive load	I leads V

💡 Ideal PF = 1 (unity) — maximum efficiency

COMPLETE POWER SUMMARY

Circuit	P (Watts)	Q (VAR)	S (VA)	PF ($\cos\phi$)	ϕ
Pure R	$V^2/R = I^2 R$	0	$= P$	1	0°
Pure L	0	VI (lagging)	$= Q$	0	90°
Pure C	0	VI (leading)	$= Q$	0	-90°
R-L Series	$I^2 R$	$I^2 X_L$	$I^2 Z$	R/Z (lag)	0° to 90°
R-C Series	$I^2 R$	$I^2 X_C$	$I^2 Z$	R/Z (lead)	0° to -90°
R-L-C Series	$I^2 R$	$I^2 (X_L - X_C)$	$I^2 Z$	R/Z	Depends on X_L, X_C

💡 **Key Insight:** $S^2 = P^2 + Q^2$ is a right triangle. P is always +ve; Q is +ve for inductive (lagging) and -ve for capacitive (leading). Power factor improvement = adding capacitors to reduce Q.

FLASHCARDS — PART 1 (Q1–Q16)

Quick Q&A for exam preparation. Cover the answer and test yourself!

Q1: State Ohm's Law

$V = IR$ — Voltage = Current $\times$ Resistance (at const. temp.)

Q3: What is KVL?

$\sum V = 0$ in a loop — Sum of voltages around a closed loop = 0

Q5: Superposition theorem applies to?

Linear bilateral circuits only — NOT for power (I^2R is non-linear)

Q7: Current source is deactivated by?

Open circuit ($I = 0$, $R_{\text{internal}} = \infty$)

Q9: Average value of sinusoidal AC?

$V_{\text{avg}} = 2V_m/\pi \approx 0.637 V_m$ — (half cycle)

Q11: Crest/Peak factor of sine wave?

$K_p = V_m/V_{\text{rms}} = \sqrt{2} \approx 1.414$

Q13: Capacitive reactance formula?

$X_C = 1/\omega C = 1/2\pi fC$ — Decreases with frequency

Q15: Phase angle in pure inductor?

$\phi = +90^\circ$ — Voltage leads current by 90°

Q2: What is KCL?

$\sum I = 0$ at a node — Sum of currents at any node = 0

Q4: Power in DC circuit?

$P = VI = I^2R = V^2/R$ — Units: Watts

Q6: Voltage source is deactivated by?

Short circuit ($V = 0$, $R_{\text{internal}} = 0$)

Q8: RMS value of AC represents?

$V_{\text{rms}} = V_m/\sqrt{2} \approx 0.707 V_m$ — Effective / heating value

Q10: Form factor of sine wave?

$K_f = V_{\text{rms}}/V_{\text{avg}} = 1.11$

Q12: Inductive reactance formula?

$X_L = \omega L = 2\pi fL$ — Increases with frequency

Q14: Impedance formula (general)?

$Z = R + jX = \sqrt{R^2 + X^2} \angle \tan^{-1}(X/R)$ — Ω

Q16: Phase angle in pure capacitor?

$\phi = -90^\circ$ — Current leads voltage by 90°

FLASHCARDS — PART 2 (Q17–Q30)

Q17: Power factor formula?

$$\cos \phi = P/S = R/Z \quad \text{— Range: 0 to 1}$$

Q19: Reactive power formula?

$$Q = VI \sin \phi \quad \text{— Unit: VAR}$$

Q21: Angular frequency formula?

$$\omega = 2\pi f = 2\pi/T \quad \text{— rad/s}$$

Q23: Two resistors in parallel?

$$R_{eq} = R_1 R_2 / (R_1 + R_2) \quad \text{— Product over sum}$$

Q25: Energy stored in inductor?

$$E = \frac{1}{2} L I^2 \quad \text{— Joules, stored in magnetic field}$$

Q27: Resonant frequency (series)?

$$f_r = 1 / (2\pi \sqrt{LC}) \quad \text{— } Z = R \text{ (min), } I = \text{max}$$

Q29: KCL is based on?

Conservation of charge — What goes in must come out

Q18: Active power formula?

$$P = VI \cos \phi \quad \text{— Unit: Watt (W)}$$

Q20: Apparent power formula?

$$S = VI = \sqrt{P^2 + Q^2} \quad \text{— Unit: VA}$$

Q22: Resistance temperature relation?

$$R_T = R_0 (1 + \alpha T) \quad \text{— } \alpha = \text{temp. coefficient}$$

Q24: Voltage divider for R_1 (series)?

$$V_1 = V \times R_1 / (R_1 + R_2)$$

Q26: Energy stored in capacitor?

$$E = \frac{1}{2} C V^2 \quad \text{— Joules, stored in electric field}$$

Q28: Power factor of pure inductor?

$$PF = 0 \text{ (lagging)} \quad \text{— } P = 0, \text{ only reactive power}$$

Q30: What does "lagging" mean in AC?

Current lags voltage — Inductive circuit ($\phi > 0^\circ$)

REACTANCE & IMPEDANCE SUMMARY

Element	Reactance	Impedance (Z)	Behavior at DC (f=0)	Behavior at High f
Resistor (R)	—	R	Normal (R)	Normal (R)
Inductor (L)	$X_L = 2\pi f L$	$jX_L = j \cdot 2\pi f L$	Short ($X_L=0$)	Open ($X_L \rightarrow \infty$)
Capacitor (C)	$X_C = 1/2\pi f C$	$-jX_C = 1/j \cdot 2\pi f C$	Open ($X_C \rightarrow \infty$)	Short ($X_C=0$)
R-L Series	X_L	$\sqrt{R^2 + X_L^2}$	R (only)	Dominantly inductive
R-C Series	X_C	$\sqrt{R^2 + X_C^2}$	Dominantly capacitive	R (only)
R-L-C Series	$X_L - X_C$	$\sqrt{R^2 + (X_L - X_C)^2}$	Capacitive (large X_C)	Inductive (large X_L)

🔧 EXAM TIPS — COMMON MISTAKES & TRICKS

⚠️ KCL & KVL SIGN CONVENTION MISTAKES

✖ **Mistake 1:** Not assigning directions before applying KCL/KVL
✔ **Fix:** Always assign current directions (assumed) and voltage polarities first. If result is negative, current actually flows opposite to assumed direction.

✖ **Mistake 2:** Inconsistent sign convention in KVL loop
✔ **Fix:** Choose loop direction (CW or CCW) and stick to it. Rise = +, Drop = -. For a resistor: if traversing from - to + = +ve; if + to - = -ve.

✖ **Mistake 3:** Applying Superposition to find power
✔ **Fix:** Superposition is only for linear quantities (I, V). Power $P = I^2R$ is NOT linear. Find total I/V first, then calculate P.

✖ **Mistake 4:** Forgetting to deactivate dependent sources
✔ **Fix:** In Superposition, only INDEPENDENT sources are deactivated. Dependent sources remain active always.

⚠️ AC CIRCUIT MISTAKES

✖ **Mistake 5:** Confusing RMS and average values
✔ **Fix:** $RMS = V_m/\sqrt{2} \approx 0.707V_m$; $Average = 2V_m/\pi \approx 0.637V_m$. RMS is for power calculations. Multimeters read RMS.

✖ **Mistake 6:** Forgetting phase angles in AC
✔ **Fix:** Always check if circuit is R, L, C or combination. Pure L $\rightarrow$ I lags V 90° ; Pure C $\rightarrow$ I leads V 90° ; R $\rightarrow$ in phase.

✖ **Mistake 7:** Adding impedances like resistances
✔ **Fix:** Impedances are complex numbers. $Z = R + jX$. Cannot just add magnitudes. Use phasor addition.

⚠️ POWER & POWER FACTOR MISTAKES

✖ **Mistake 8:** Confusing P, Q, and S
✔ **Fix:** P (Watts) = real/useful; Q (VAR) = reactive; S (VA) = total/apparent. $S = \sqrt{P^2 + Q^2}$, NOT $P + Q$.

✖ **Mistake 9:** Saying "power factor = $\cos\phi$ " without specifying leading/lagging
✔ **Fix:** Always specify: PF = 0.8 lagging (inductive) or PF = 0.9 leading (capacitive). Lagging = I lags V.

✖ **Mistake 10:** Forgetting units
✔ **Fix:** Memorize all units: R $\rightarrow\Omega$, V $\rightarrow$ V, I $\rightarrow$ A, f $\rightarrow$ Hz, L $\rightarrow$ H, C $\rightarrow$ F, P $\rightarrow$ W, Q $\rightarrow$ VAR, S $\rightarrow$ VA, X $\rightarrow\Omega$, Z $\rightarrow\Omega$, $\omega\rightarrow$ rad/s.

💡 EXAM TRICKS & MEMORY AIDS

🔵 **ELI the ICE man:** In inductor (L), E (voltage) leads I (current). In capacitor (C), I (current) leads E (voltage). Mnemonic: **ELI the ICE man**

🟢 **Series vs Parallel:** Series $\rightarrow$ same I; Parallel $\rightarrow$ same V. $R_{eq}(\text{series}) >$ any R; $R_{eq}(\text{parallel}) <$ smallest R.

🟡 **Power Triangle:** S is hypotenuse (longest side). $P = S \cos\phi$; $Q = S \sin\phi$. For unity PF: $S = P$, $Q = 0$.

🔴 **Form factor always ≥ 1 :** $K_f = RMS/Avg \geq 1$ always. For pure DC, $K_f = 1$. For sine, $K_f = 1.11$.

🟣 **Resonance:** At series resonance — Z minimum, I maximum, PF = 1, $V_L = V_C$ (can be $> V_{supply}$!).

⚡ **Capacitor at DC:** Open circuit (blocks DC). Inductor at DC: Short circuit (passes DC). Remember: capacitor charges to supply voltage; inductor carries constant current.

QUICK REVISION SHEET — ALL FORMULAS AT A GLANCE

Last-minute revision — formulas only, no explanations

⚡ DC CIRCUIT FORMULAS

Ohm's Law
 $V = IR$

Power (DC)
 $P = VI = I^2R = V^2/R$

Energy
 $W = Pt = Vit \text{ [J]}$

Series Req
 $R_1 + R_2 + R_3 + \dots$

Parallel Req
 $1 / (1/R_1 + 1/R_2 + 1/R_3)$

2 Parallel R
 $R_1 R_2 / (R_1 + R_2)$

Voltage Divider
 $V_1 = V \cdot R_1 / (R_1 + R_2)$

Current Divider
 $I_1 = I \cdot R_2 / (R_1 + R_2)$

Conductance
 $G = 1/R \text{ [S]}$

KCL
 $\Sigma I = 0 \text{ (node)}$

KVL
 $\Sigma V = 0 \text{ (loop)}$

Resistivity
 $R = \rho L/A$

🔌 AC FUNDAMENTALS

AC Voltage
 $v = V_m \sin(\omega t + \phi)$

Ang. Freq.
 $\omega = 2\pi f \text{ [rad/s]}$

Frequency
 $f = 1/T \text{ [Hz]}$

RMS Value
 $V_{rms} = V_m / \sqrt{2} = 0.707 V_m$

Average Value
 $V_{avg} = 2V_m / \pi = 0.637 V_m$

Form Factor
 $K_f = V_{rms} / V_{avg} = 1.11$

Crest Factor
 $K_p = V_m / V_{rms} = \sqrt{2} = 1.414$

Peak-to-Peak
 $V_{pp} = 2V_m$

Period
 $T = 1/f = 2\pi / \omega \text{ [s]}$

📐 AC CIRCUITS & IMPEDANCE

Inductive React.
 $X_L = \omega L = 2\pi fL$

Capacitive React.
 $X_C = 1/\omega C = 1/2\pi fC$

Impedance Mag.
 $|Z| = \sqrt{R^2 + X^2}$

Phase Angle
 $\phi = \tan^{-1}(X/R)$

Z Complex
 $Z = R + jX$

Z (R-L)
 $\sqrt{R^2 + X_L^2} \angle +\phi$

Z (R-C)
 $\sqrt{R^2 + X_C^2} \angle -\phi$

Z (R-L-C)
 $\sqrt{R^2 + (X_L - X_C)^2}$

Resonant Freq.
 $f_r = 1/(2\pi\sqrt{LC})$

Pure R phase
 $\phi = 0^\circ \text{ (in phase)}$

Pure L phase
 $\phi = +90^\circ \text{ (V leads I)}$

Pure C phase
 $\phi = -90^\circ \text{ (I leads V)}$

⚡ POWER FORMULAS

Active Power P
 $P = VI \cos\phi = I^2R \text{ [W]}$

Reactive Power Q
 $Q = VI \sin\phi = I^2X \text{ [VAR]}$

Apparent Power S
 $S = VI = \sqrt{P^2 + Q^2} \text{ [VA]}$

Power Factor
 $\cos\phi = P/S = R/Z$

Power Triangle
 $S^2 = P^2 + Q^2$

$\tan\phi$
 $\tan\phi = Q/P = X/R$

$\sin\phi$
 $\sin\phi = Q/S = X/Z$

Energy (L)
 $E = \frac{1}{2}LI^2 \text{ [J]}$

Energy (C)
 $E = \frac{1}{2}CV^2 \text{ [J]}$

🔧 UNITS QUICK REFERENCE

Quantity	Symbol	Unit	Quantity	Symbol	Unit	Quantity	Symbol	Unit
Voltage	V	Volt (V)	Frequency	f	Hertz (Hz)	Active Power	P	Watt (W)
Current	I	Ampere (A)	Ang. Freq.	ω	rad/s	Reactive Power	Q	VAR

Quantity	Symbol	Unit	Quantity	Symbol	Unit	Quantity	Symbol	Unit
Resistance	R	Ohm (Ω)	Inductance	L	Henry (H)	Apparent Power	S	VA
Impedance	Z	Ohm (Ω)	Capacitance	C	Farad (F)	Energy	W	Joule (J)
Reactance	X	Ohm (Ω)	Time Period	T	Second (s)	Conductance	G	Siemens (S)

 **ALL THE BEST FOR YOUR EXAM! YOU'VE GOT THIS!** 
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